

Phone Calls & WhatsApp Voice Notes

SIH26097 · How the channels work, what they cost, and what we do differently · 25 Sept 2026

The problem statement asks for three ways in: **a phone call for people with basic phones, WhatsApp voice notes**, and **a simple app or kiosk**. This note covers the first two — the ones that need an outside company and cost money per use.

1 · How a phone call actually works

You rent a phone number from an Indian telecom company. When someone calls it, the company doesn't play a recording — it opens a live pipe to our server and streams the caller's voice across as they speak. We send audio back down the same pipe. The caller just hears a conversation.

Beneficiary dials a toll-free number on a ₹800 keypad phone

↓ telecom company streams their voice to us, live

↓ we turn speech into text · work out what they said · pick the reply

↓ we send audio back down the same line

↓ they hear an answer – about 1.5 seconds later

Two Indian companies matter here. **Exotel** is our choice; it bills in rupees, its numbers are cleared with Indian telecom regulators, and the audio stays inside India. **Twilio** is American, costs more, and routes audio abroad — which is awkward when the caller is a government scheme beneficiary.

Why it matters that it's a phone call, not an app: it uses the voice network, not mobile data. It works on a phone with no internet, no screen worth reading, and no app store. That is most of the people this scheme is for.

2 · How WhatsApp voice notes actually work

Someone records a voice note and sends it to our number. WhatsApp tells our server a message arrived; we fetch the audio file, listen to it, and send a reply. The reply can be text or, if done properly, **another voice note** — which matters, because someone who cannot read a form usually cannot read our answer either.

Beneficiary holds the mic button and speaks

↓ WhatsApp tells our server a voice note arrived

↓ we download it · turn it into text · work out the reply

↓ we send back a voice note they can just listen to

There are two ways to connect. Going **direct to Meta** is cheaper and lets you send voice replies. Going **through Twilio** is quicker to set up but adds a markup on every message and makes voice replies harder, which pushes you towards text-only replies.

Deadline — 1 October 2026, six days away. Right now, replying to someone who messaged you first is free. From 1 October, Meta charges for those replies after the first **1,000 per month**. Worse: **any account without a payment method saved by 30 September stops being able to send these replies at all**.

3 · Which one is better

	Phone call (Exotel)	WhatsApp voice notes	Offline kiosk / app
Reaches a basic keypad phone	Yes	No	No
Needs mobile internet	No	Yes	No
Works on a shared family phone	Yes — we can call back	Poorly — messages sit in someone else's chat	Yes
Feels like a conversation	Yes	No — send, wait, reply	Yes
Can pause and come back later	Only if built for it	Yes, naturally	Yes
Leaves proof for the file	Yes	Yes — the chat is the record	Yes
Cost per person	Middle	Low	Nearly zero
Set-up difficulty	Hardest	Medium	Easy

Verdict: they are not competing, they are for different people. The phone call is the only one that reaches the poorest beneficiary and the only one that reaches a woman on a shared handset. WhatsApp is cheapest and leaves a natural paper trail. The offline kiosk costs nothing to run and is the only thing that works when the network is down. We need all three, running off one shared brain — not three separate products.

4 · What it costs

Worked on **1,000 beneficiaries, a four-minute interview each**. Vendor rates are current published figures and should be re-checked before any number goes in front of a judge.

Route	Per person	Per 1,000	What drives the cost
Phone call — Exotel + paid speech, live voice replies	≈ ₹7.85	≈ ₹7,850	Call minutes (₹0.80–1.00/min) plus speech charges
Phone call — Exotel + pre-recorded questions	≈ ₹5.60	≈ ₹5,600	Recording the questions once removes most of the speech bill
WhatsApp — direct to Meta, <i>today</i>	≈ ₹2.00	≈ ₹2,000	Replies still free; only speech costs money
WhatsApp — direct to Meta, after 1 Oct	≈ ₹3.10	≈ ₹3,080	≈10 replies each; first 1,000 a month free, then ≈₹0.12 each
WhatsApp via Twilio	higher	higher	Same as above plus Twilio's cut on every message
Offline kiosk — our own speech software	≈ ₹0	≈ ₹0	No call, no data, no vendor. Cost is the device, once

Three things that fall straight out of this table.

1. **Recording the questions once saves about 29% of every call.** The questions never change — there is no reason to pay a computer to re-read them 1,000 times. It is also faster, which is what stops a caller hanging up.
2. **Twilio is the expensive choice on both channels.** Exotel and going direct to Meta are cheaper and keep the data in India.
3. **The offline route is free per use.** For a scheme with a fixed budget and a hard administrative spending cap, that is not a small detail.

5 · What we do differently

One brain, four doors. We build the interview once, then plug the phone line, WhatsApp, the kiosk and the field-worker's phone into it. Adding a channel becomes a week, not a rewrite — and the questions can never drift apart between channels.

1. **Reply by voice on WhatsApp, not by text.** If the person could read our answer, they could have filled in the form.
2. **Record the questions once, in a real human voice.** Cheaper, roughly a second faster per turn, works with the network off — and someone can actually listen to all forty prompts and fix the ones that sound cold. The problem statement asks for "empathetic, not administrative"; that is achieved by ear, not by software.
3. **Survive a dropped call.** Rural calls drop constantly. We save each answer as it is confirmed, so calling back resumes where it stopped instead of starting the interview again.
4. **Say so when we cannot hear them.** Speech software gets roughly one word in four wrong on a village phone line — and that is the *best* available; Google's gets three in five wrong. Rather than trust whatever text comes back, we check our own confidence and simply ask again, which is what a considerate human interviewer does anyway.
5. **A third door: works with no internet at all.** Our own speech software on a cheap phone at the panchayat office. Costs nothing per interview and can be demonstrated with the phone in aeroplane mode.
6. **A fourth door reality demands: the field worker's phone.** Half of rural women over 15 own no phone, and the scheme requires at least 30% women in every training batch. Without a mode the ASHA or Anganwadi worker can use on the doorstep, that target cannot be met.
7. **Not built on a price that expires next week.** Our costing assumes WhatsApp replies are paid, because from 1 October they are. Our cheapest channel is the one with no vendor in it at all.
8. **A phone number people ring, not one that rings them.** An incoming toll-free line sidesteps the regulatory questions around automated outbound calling, and it puts the beneficiary in control of when the conversation happens — which on a

shared handset is the whole game.

Sources: Exotel and Twilio published India rates; Meta WhatsApp Business pricing including the 1 Oct 2026 change; Sarvam published API rates; AI4Bharat speech accuracy benchmark (Vistaar). Vendor prices move — re-check before publishing any figure.